

Solutions to Monthly Online Test - 1 | JEE - 2022

PHYSICS

1.(B) $W = \vec{F} \cdot \vec{S} = 9 + 16 = 25J$

2.(B) $\sqrt{(0.5)^2 + (0.8)^2 + c^2} = 1$

3.(B) $(2\hat{i} + 3\hat{j} + 8\hat{k}) \cdot (4\hat{i} - 4\hat{j} + \alpha\hat{k}) = 0$

$\therefore 8 - 12 + 8\alpha = 0 \quad \therefore \alpha = \frac{1}{2}$

4.(C) $\vec{A} \cdot \vec{B} = 9 + 16 - 25 = 0$

$\therefore$ Angle between A and B is 90°

5.(B) $\vec{R} = \vec{A} + \vec{B} = 12\hat{i} + 5\hat{j}$

$R = |\vec{R}| = \sqrt{(12)^2 + (5)^2} = 13; \quad \hat{R} = \frac{\vec{R}}{|\vec{R}|}$

6.(A) Component of A along B

$= A \cos \theta$

$= \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} = \frac{2 + 3}{\sqrt{1+1}} = \frac{5}{\sqrt{2}}$

7.(A) $|\vec{A} \times \vec{B}| = \sqrt{3} \vec{A} \cdot \vec{B}$

$\Rightarrow AB \sin \theta = \sqrt{3} AB \cos \theta \quad \text{or} \quad \tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ$

$|\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB \cos 60^\circ} = \sqrt{A^2 + B^2 + AB}$

8.(A) One vector will be perpendicular to the cross-product of other two vectors.

$(5\hat{i} + n\hat{j} + \hat{k})$ will be perpendicular to $(2\hat{i} + 3\hat{j} - 2\hat{k}) \times (-\hat{i} + 2\hat{j} + 3\hat{k})$

$\Rightarrow (5\hat{i} + n\hat{j} + \hat{k}) \cdot ((2\hat{i} + 3\hat{j} - 2\hat{k}) \times (-\hat{i} + 2\hat{j} + 3\hat{k})) = 0$

$\Rightarrow (5\hat{i} + n\hat{j} + \hat{k}) \cdot (13\hat{i} - 4\hat{j} + 7\hat{k}) = 0$

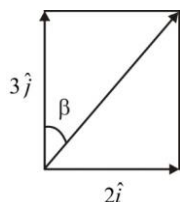
$\Rightarrow 65 - 4n + 7 = 0 \quad \Rightarrow \quad n = 18$

9.(C) $\vec{A} \cdot \vec{B} = 0 \quad \therefore \vec{A} \perp \vec{B}$

$\vec{A} \cdot \vec{C} = 0 \quad \therefore \vec{A} \perp \vec{C}$

$\vec{A}$ is perpendicular to both $\vec{B}$ and $\vec{C}$ and $\vec{B} \times \vec{C}$ is also perpendicular to both $\vec{B}$ and $\vec{C}$. Therefore, $\vec{A}$ is parallel to $\vec{B} \times \vec{C}$.

10.(C)



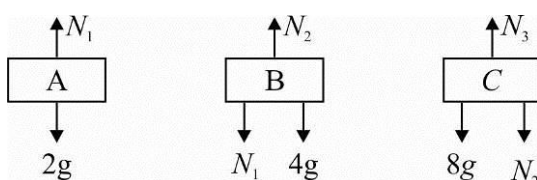
$$\tan \beta = \frac{2}{3} \text{ or } \beta = \tan^{-1}\left(\frac{2}{3}\right)$$

11.(C) $\vec{A} + \vec{B}$ will be in the plane containing $\vec{A}$ and $\vec{B}$, whereas $\vec{A} \times \vec{B}$ will be perpendicular to that plane.

12.(B) Area of parallelogram: $|\vec{A} \times \vec{B}| = AB/2$ (given)

$$\Rightarrow AB \sin \theta = AB/2 \Rightarrow \sin \theta = 1/2 \Rightarrow \theta = 30^\circ$$

13.(D)



$$N_1 = 2g \quad N_2 = N_1 + 4g = 6g \quad \therefore \text{force applied by C on B} = 6g$$

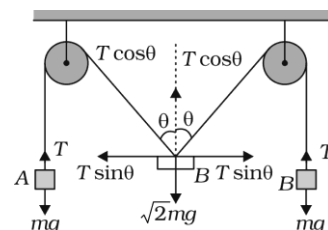
14.(C)

For equilibrium in vertical direction for body B we have

$$\sqrt{2} mg = 2T \cos \theta = 2(mg) \cos \theta$$

($\because T = mg$ at equilibrium)

$$\therefore \cos \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$



15.(A)

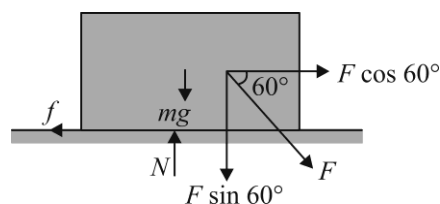
The forces acting on the block are shown. Since the block is not moving forward for the maximum force F applied

$$F \cos 60^\circ = f = \mu N \quad \dots (i)$$

$$\text{And } F \sin 60^\circ + mg = N \quad \dots (ii)$$

$$\text{From, (i) and (ii), } F \cos 60^\circ = \mu [F \sin 60^\circ + mg]$$

$$\Rightarrow F = \frac{\mu mg}{\cos 60^\circ - \mu \sin 60^\circ} = \frac{\frac{1}{2\sqrt{3}} \times \sqrt{3} \times 10}{\frac{1}{2} - \frac{1}{2\sqrt{3}} \times \frac{\sqrt{3}}{2}} = \frac{5}{\frac{1}{4}} = 20 \text{ N}$$



16.(D) $T_A = 10g = 100 \text{ N}$

$$T_B \cos 30^\circ = T_A \therefore T_B = \frac{200}{\sqrt{3}} \text{ N}$$

$$T_B \sin 30^\circ = T_C \therefore T_C = \frac{100}{\sqrt{3}} \text{ N}$$

17.(A) Only two forces are acting, mg and net contact force (resultant of friction and normal reaction) from the inclined plane. Since the body is at rest. Therefore these two forces should be equal and opposite.

18.(C) Given horizontal force $F = 25 \text{ N}$ and coefficient of friction between block and wall $(\mu) = 0.4$

We know that at equilibrium horizontal force provides the normal reaction to the block against the wall.

Therefore, normal reaction to the block $(R) = F = 25 \text{ N}$.

We also know that weight of the block $(W) = \text{Frictional force}$

$$= \mu R = 0.4 \times 25 = 10 \text{ N}.$$

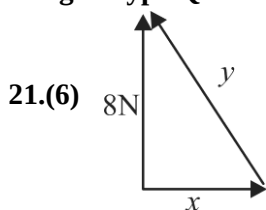
19.(B) If $\vec{A}$ is perpendicular to $\vec{B}$, then $\vec{A} \cdot \vec{B} = 0$ and $\vec{A} \times \vec{B} \neq 0$

20.(C) $\vec{S} = 75\hat{j} + [60\cos 45^\circ \hat{j} - 60\sin 45^\circ \hat{i}] + 20\hat{i}$

$$= (20 - 60 \times 0.7)\hat{i} + (60 \times 0.7 + 75)\hat{j} = -22\hat{i} + 117\hat{j}$$

$$S = \sqrt{22^2 + 117^2} = \sqrt{484 + 13689} = \sqrt{14173} = 119 \text{ km}$$

Integer Type Questions



$$x + y = 16$$

$$\text{Also, } y^2 = 8^2 + x^2$$

$$\text{or } y^2 = 64 + (16 - y)^2 \quad [\because x = 16 - y]$$

$$\text{or } y^2 = 64 + 256 + y^2 - 32y \quad \text{or } y = 10 \text{ N}$$

$$\therefore x + 10 = 16 \quad \text{or } x = 6 \text{ N}$$

22.(120) Given $R = A = B$, it will give $\theta = 120^\circ$

23.(90) $a^2 + b^2 + 2ab\cos\theta = a^2 + b^2 - 2ab\cos\theta$

$$\text{or } 4ab\cos\theta = 0$$

$$\text{but } 4ab \neq 0 \Rightarrow \cos\theta = 0 \quad \text{or } \theta = 90^\circ$$

24.(60) $\vec{A} + \vec{B} = \vec{C}$

$$(\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B}) = \vec{C} \cdot \vec{C}$$

$$\Rightarrow A^2 + B^2 + 2AB\cos\theta = C^2$$

$$4^2 + 5^2 + 2 \times 4 \times 5 \cos\theta = 61$$

$$\Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

25.(2) $A + B = 7$ and $A - B = 3$

$$\therefore A = 5 \quad \text{and } B = 2$$

CHEMISTRY

1.(D) Statement (D) is a postulate for Bohr's atomic model.

2.(B) $E = h\nu \Rightarrow E = \frac{hc}{\lambda} \Rightarrow E = hc\bar{\nu}$

3.(C) Wavelength increase in the order: X-ray < UV rays < Microwaves < Radio waves.

4.(B) $E = h\nu = 6.626 \times 10^{-34} \times 6 \times 10^{14} = 3.9 \times 10^{-19} \text{ J}$

5.(A) Work function $= \frac{hc}{\lambda_0} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{500 \times 10^{-9}} \text{ J} \approx 4 \times 10^{-19} \text{ J}$

6.(B) According to Bohr's theory, $r_n = \frac{n^2 h^2}{4\pi^2 K Z m e^2}$

7.(A) $v = \frac{2\pi K Z e^2}{nh}$; for H-atom, $Z = 1, n = 1$

$$v = \frac{2\pi K e^2}{h}$$

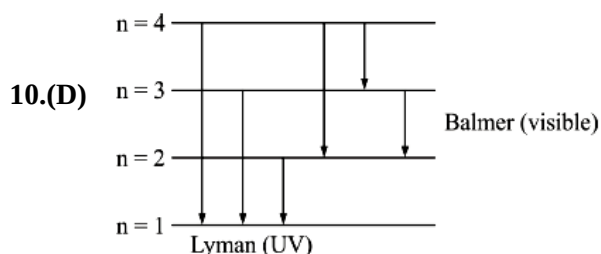
So, 2nd orbit of He^+ , $Z = 2, n = 2$

$$v' = \frac{2\pi K e^2 \times 2}{2h} = \frac{2\pi K e^2}{h} = v$$

8.(C) $n = 5$ to $n = \infty$

$$\Delta E = 13.6 \times 1^2 \left[\frac{1}{5^2} - \frac{1}{\infty^2} \right] \text{ eV} = 13.6 \times \frac{1}{25} \text{ eV} = 0.544 \text{ eV}$$

9.(A) Angular momentum $= \frac{nh}{2\pi} = \frac{2 \times h}{2\pi} = 2.11 \times 10^{-34} \text{ J sec}$ (first excited state, $n = 2$)



11.(B) It could explain the spectra of atoms/ions having only one electron and not two or more electrons.

12.(B) $\Delta x \cdot \Delta p = \frac{h}{4\pi}$; $(\Delta p)^2 = \frac{h}{4\pi}$

$$(m\Delta v)^2 = \frac{h}{4\pi}; \quad \Delta v = \frac{1}{2m} \sqrt{\frac{h}{\pi}}$$

13.(D) $\lambda = \frac{h}{\sqrt{2 \times K.E. \times m}} \therefore \frac{\lambda_m}{\lambda_{2m}} = \frac{\sqrt{2}}{1}$

14.(A) m_ℓ can be only from $-\ell$ to $+\ell$ $\therefore m_\ell \neq -2$ when $\ell = 1$

15.(A) Orbital angular momentum = $\sqrt{\ell(\ell+1)} \frac{h}{2\pi}$

16.(C) For H-atom, energy of an orbital solely depends on n
 $1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f < \dots$

17.(B)

	Number of radial nodes ($n - \ell - 1$)	Number of angular nodes (ℓ)
2p	0	1
3p	1	1

18.(C) The outer most electronic configuration of V is $[\text{Ar}] 3d^3 4s^2$ $\therefore V^{2+}$ is $[\text{Ar}] 3d^3 4s^0$

19.(C) Number of waves made by electron in one complete revolution in an orbit around nucleus are equal to principal quantum number of the orbit. In this case orbit is third orbit hence number of waves made by electron are three

20.(C) Total electrons in 3d-subshell = 10, but electrons with $s = +\frac{1}{2}$ is equal to 5.

Integer Type Questions

21.(8) $\text{Cr}(Z=24) : 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$

$n=3, \ell=0 \Rightarrow 3s$ (No. of $e^- = 2$)

$n=2, \ell=1 \Rightarrow 2p$ (No. of $e^- = 6$)

Total no. of $e^- = 8$

22.(3) $5 \rightarrow 2, 4 \rightarrow 2, 3 \rightarrow 2$

Total spectral lines in visible region = 3

23.(8) $\bar{\nu} = \frac{1}{\lambda} = R_H \times Z^2 \left[\frac{1}{n_i^2} - \frac{1}{n_f^2} \right]$

$n_i = 1, n_f = 3$

$\bar{\nu} = R_H \times 9 \left[1 - \frac{1}{9} \right]; \quad \bar{\nu} = R_H \times 8$

24.(16) Degeneracy of principal energy level = n^2

For fourth energy level $n = 4$ hence its degeneracy is equal to 16.

25.(2) $r_n = 0.529 \times \frac{n^2}{Z} \text{ \AA}$

Radius of 1st Bohr's orbit of hydrogen atom = $0.529 \text{ \AA}$

For Be^{3+} , $\frac{0.529 \times n^2}{4} = 0.529$

$n^2 = 4; \quad n = 2$

MATHEMATICS

1.(A) $\alpha + \beta = \frac{-b}{a}$

$\therefore a\alpha + b = -a\beta$ and $a\beta + b = -a\alpha$

$\therefore S = \frac{-1}{a} \left(\frac{\alpha + \beta}{\alpha\beta} \right) = \frac{-1}{a} \left(\frac{-b/a}{c/a} \right) = \frac{b}{ac}$

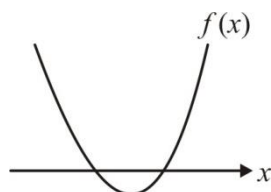
2.(D) $a > 0; f(0) = c > 0; \frac{-b}{a} > 0$

$\therefore -b > 0 \Rightarrow b < 0$

Hence $abc < 0$ and $D = b^2 - 4ac > 0$

3.(A) $P(x) = x^2 - (2 - P)x + (P - 2)$

We must have $D > 0$



$\Rightarrow (P - 2)(P - 6) > 0$

$\Rightarrow P \in (-\infty, 2) \cup (6, \infty)$

4.(D) $111x^3 - 11x - 1 = 0 \quad \therefore \quad \alpha + \beta + \gamma = 0$

$\Sigma\alpha\beta = \frac{-11}{111}, \alpha\beta\gamma = \frac{1}{111} \Rightarrow \frac{1}{(\alpha\beta)^2} + \frac{1}{(\beta\gamma)^2} + \frac{1}{(\gamma\alpha)^2}$

$= \frac{\alpha^2 + \beta^2 + \gamma^2}{(\alpha\beta\gamma)^2} = \frac{(\alpha + \beta + \gamma)^2 - 2\Sigma\alpha\beta}{(\alpha\beta\gamma)^2} = 2442$

5.(A) $f(0) < 0 \Rightarrow k^2 - 3k + 4 < 0$

which is a quadratic in k

and $D < 0 \therefore k^2 - 3k + 4 > 0 \quad \forall k \in R.$

$f(0) < 0$ not possible.

6.(A) The given equation may be written as

$x^2 - px - (p + c) = 0 \quad \therefore \quad \alpha + \beta = p \text{ and } \alpha\beta = -(p + c)$

Now $(\alpha + 1)(\beta + 1) = \alpha\beta + \alpha + \beta + 1$

$= -p - c + p + 1 = 1 - c$

7.(C) $\alpha(\alpha\beta)^5 = 32 \Rightarrow \alpha = 1 \quad [\because \alpha\beta = 2]$

The value of α will satisfy the given equation $1 + a + 2 = 0 \Rightarrow a = -3$

8.(B) $\therefore f(2)$ is minimum
 $\therefore f(x)$ will be symmetric about $x=2$
 $\therefore$ greater is magnitude of difference between value of x and 2, greater is the value of $f(x)$
 $\therefore f(2) < f(0) < f(5)$

9.(B) $lx^2 + mx + n = 0$; $px^2 + qx + r = 0$
 Root x_1, x_2 ; roots x_3, x_4
 As $|x_1 - x_2| = |x_3 - x_4|$
 $\Rightarrow (x_1 + x_2)^2 - 4x_1x_2 = (x_3 + x_4)^2 - 4x_3x_4 \Rightarrow \frac{D_1}{D_2} = \frac{l^2}{p^2}$

10.(A) $D < 0$
 $1 - 4m < 0 \Rightarrow 4m > 1; m > \frac{1}{4}$

11.(B) $D < 0$
 $b^2 - 4ac < 0; b^2 < 4ac$

12.(B) With the condition, $c^2 = 4d$, then equation $x^2 - cx + d = 0$
 has equal roots. Each root $= \frac{c}{2}$.
 As $\frac{c}{2}$ is a common root. $\therefore$ it is a root of $x^2 - ax + b = 0$
 $\left(\frac{c}{2}\right)^2 - a\left(\frac{c}{2}\right) + b = 0 \Rightarrow \frac{4d}{4} - \frac{ac}{2} + b = 0 \Rightarrow b + d = \frac{ac}{2}$

13.(C) $m = \frac{-b}{2a} = 1, n = \frac{-D}{4a} = 4$
 Now check alternative for parabola having vertex $\left(2m, \frac{n}{2}\right)$ i.e., $(2, 2)$

14.(C) It is obvious m and n are the roots of the equation $x^2 - 4x - 4 = 0$
 $\Rightarrow m + n = 4$ and $mn = -4$
 Now $m^2 + n^2 + 4mn = (m + n)^2 + 2mn = 4^2 + 2(-4) = 16 - 8 = 8$

15.(B) $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ $\begin{matrix} \alpha \\ 2\alpha \end{matrix}$
 $3\alpha = \frac{-(3a-1)}{a^2-5a+3}; 2\alpha^2 = \frac{2}{(a^2-5a+3)}$
 $\frac{(3a-1)^2}{9(a^2-5a+3)^2} = \frac{1}{(a^2-5a+3)}$
 $9a^2 - 6a + 1 = 9a^2 - 45a + 27$
 $39a = 26; \boxed{a = \frac{2}{3}}$

16.(A) Student 1 : $x^2 - 5x + 6 = 0$

Student 2 : $x^2 - 7x + 10 = 0$

Correct equation $x^2 - 7x + 6 = (x-1)(x-6) = 0$, $x = 1, 6$

17.(A) $x^2 - 2x + 3 = 0$ $\begin{matrix} \alpha \\ \beta \end{matrix}$

$$\frac{\alpha-1}{\alpha+1} = y$$

$$\alpha - 1 = \alpha y + y$$

$$\alpha = \frac{1+y}{1-y}$$

$$\frac{(1+y)^2}{(1-y)^2} - 2\left(\frac{1+y}{1-y}\right) + 3 = 0$$

$$y^2 + 2y + 1 - 2(1 - y^2) + 3(1 + y^2 - 2y) = 0$$

$$6y^2 - 4y + 2 = 0$$

$$3y^2 - 2y + 1 = 0$$

18.(B) We have $(x+1)^2 > 5x - 1 \Rightarrow x^2 - 3x + 2 > 0$

$$\Rightarrow x < 1 \text{ or } x > 2 \quad \dots(1)$$

$$(x+1)^2 < 7x - 3 \Rightarrow x^2 - 5x + 4 < 0 \Rightarrow 1 < x < 4 \quad \dots(2)$$

$$(1) \text{ and } (2) \quad x \in (2, 4)$$

$\therefore$ Only one integral value of x which is 3.

19.(C) $x^2 - 11x + a = 0$; $x^2 - 14x + 2a = 0$

$$3x = a; \quad x = \frac{a}{3}$$

$$\frac{a^2}{9} - \frac{11a}{3} + a = 0 \Rightarrow a(a - 24) = 0 \Rightarrow a = 0, 24$$

$$\text{Sum} = 24$$

20.(D) $\beta > 1$ and $\alpha < -1 \Rightarrow \frac{c}{a} < -1$

$$\text{Since } a < 0 \Rightarrow c > 0$$

$$\text{Also, } \alpha + \beta > 0 \Rightarrow -\frac{b}{a} > 0 \Rightarrow b > 0 \Rightarrow abc < 0$$

$$\text{if } x \in (0, \beta) \text{ graph lies above x-axis} \Rightarrow f(x) > 0$$

$$\text{Also } \beta > 1 \text{ and } \alpha < -1 \Rightarrow |\beta - \alpha| > 2$$

Integer Type Questions

$$21.(0) \quad x^3 + 2x^2 + 2x + c = (x^2 + bx + b)\left(x + \frac{c}{b}\right)$$

Comparing both sides each coefficient of x .

$$b + \frac{c}{b} = 2 \text{ and } b + c = 2 \Rightarrow b = c = 1$$

$$22.(4) \quad \frac{2}{\frac{1}{\alpha} + \frac{1}{\beta}} = \frac{2\alpha\beta}{\alpha + \beta} = \frac{\frac{2(8+2\sqrt{5})}{(5+\sqrt{2})}}{\frac{4+\sqrt{5}}{(5+\sqrt{2})}} = 4$$

$$23.(8) \quad 2x^2 - 35x + 2 = 0 \begin{cases} \alpha \\ \beta \end{cases}$$

$$2\alpha - 35 = \frac{-2}{\alpha} \text{ and } 2\beta - 35 = \frac{-2}{\beta}$$

$$\sqrt{\left(\frac{-2}{\alpha}\right)^3 \left(\frac{-2}{\beta}\right)^3} = \sqrt{\frac{64}{(\alpha\beta)^3}} = 8$$

$$24.(6) \quad x^2 - 8x - (n^2 - 10n) = 0$$

$$D < 0 \Rightarrow n^2 - 10n + 16 < 0 \Rightarrow 2 < n < 8, n \neq 10$$

$$25.(0) \quad y = \frac{x^2 - 2x + 2}{2x - 2} \Rightarrow x^2 - 2x + 2 = 2xy - 2y$$

$$\Rightarrow x^2 - 2x(y+1) + 2(y+1) = 0$$

as $x \in R$, so $D \geq 0$

$$(y+1)^2 - 2(y+1) \geq 0 \Rightarrow (y+1)(y-1) \geq 0$$

$$\Rightarrow y \geq 1 \text{ or } y \leq -1$$